ANALYSTLAB AFRICA

Data Science Internship Programme

Heart Disease Prediction
Business Understanding Report

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

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Week 2 Submission

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1. Introduction

Cardiovascular disease is a major health problem. The World Health Organisation reports that cardiovascular diseases are the leading cause of death globally and identifies factors including raised blood pressure, raised blood glucose and abnormal blood lipids among important cardiovascular risk factors (World Health Organisation [WHO], 2025).

The purpose of this project is to prepare a Heart Disease Prediction dataset for future machine learning analysis. The Week 2 assignment focuses on transforming raw data into a clean, structured and machine-learning-ready dataset through data inspection, cleaning, feature engineering, encoding, outlier treatment, scaling and feature selection. The dataset compiles established clinical predictor variables associated with heart disease outcomes (fedesoriano, 2021).

2. Business Problem

Healthcare organisations have access to different types of patient information, including demographic characteristics, symptoms, blood pressure measurements, cholesterol levels and cardiac examination results. Analysing these variables can help identify patterns associated with the presence of heart disease. Variables of this kind are widely reported among the key clinical risk indicators for heart disease (Centers for Disease Control and Prevention [CDC], 2025).

The business problem addressed by this project is how patient information can be prepared and analysed so that it can subsequently be used to predict whether an individual is likely to have heart disease.

The raw dataset may contain data-quality issues that can affect subsequent analysis. These include missing or invalid values, duplicate records, inconsistent data types, categorical variables and potential outliers. During data inspection, zero values were identified in variables such as RestingBP and Cholesterol. Because these variables represent clinical measurements, such values require investigation and appropriate treatment before predictive modelling.

3. Project Objective

The main objective of this project is to prepare the Heart Disease Prediction dataset for machine learning.

The analysis will specifically focus on:

- Understanding the structure and characteristics of the dataset.
- Identifying and treating missing or invalid values.
- Detecting and removing duplicate records where appropriate.
- Checking and correcting data types.
- Identifying relevant and potentially irrelevant features.
- Creating useful features where appropriate.
- Encoding categorical variables using suitable techniques.
- Detecting and treating outliers.
- Scaling or normalising numerical variables.
- Examining correlations between features.
- Selecting features based on analytical evidence.
- Producing a final machine-learning-ready dataset.

Standard preprocessing techniques such as categorical encoding and feature scaling are widely used to prepare structured clinical datasets for machine learning workflows (Pedregosa et al., 2011).

4. Target Variable

The target variable for the project is HeartDisease. It represents the outcome that a future predictive model would attempt to predict.

The remaining variables will be treated as potential predictor variables, including Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak and ST_Slope.

The project will not assume that all predictor variables are equally useful. Feature selection will be performed using analytical evidence such as correlation analysis and feature importance.

5. Importance of Predictive Modelling

Predictive modelling is important because it can use historical data to identify patterns and estimate the likelihood of future or previously unseen outcomes. Cardiovascular risk prediction models demonstrate how combinations of demographic and clinical risk factors can be used to estimate cardiovascular risk (WHO CVD Risk Chart Working Group, 2019). Evidence from applied machine learning research also indicates that appropriately engineered and preprocessed clinical features can meaningfully improve predictive model performance (Bhatt et al., 2023).

For this project, predictive modelling could potentially identify patterns in patient characteristics associated with the presence of heart disease. This could provide a basis for further investigation and support data-driven healthcare analysis.

However, predictive modelling should not be interpreted as automatically providing a medical diagnosis. A future model would require appropriate testing, validation and clinical oversight before it could be considered for real-world healthcare use.

6. Expected Business Impact

The expected business value of this project is to establish a reliable and machine-learning-ready dataset that can serve as a foundation for future predictive analysis.

A well-prepared dataset could support the future development of analytical tools that identify patient characteristics associated with heart disease. Such tools could potentially help qualified healthcare professionals identify individuals who may require further assessment or investigation. WHO notes that identifying people at higher cardiovascular risk and providing appropriate treatment can help prevent premature deaths (WHO, 2025).

The project could therefore provide value by:

- Supporting data-driven analysis of heart disease.
- Identifying variables that may contain useful predictive information.
- Providing a structured dataset for future machine learning development.
- Improving the consistency and quality of the available patient data.
- Supporting future research into cardiovascular risk prediction.
- Demonstrating a systematic approach to healthcare data preprocessing.

This Week 2 assignment is focused on data preparation rather than developing or deploying a clinical diagnostic system. Any future healthcare application would require additional validation, appropriate governance and professional clinical oversight.

7. Conclusion

Heart disease presents an important healthcare challenge, and predictive analytics provides an opportunity to analyse patient information and identify patterns associated with cardiovascular risk. The objective of this project is therefore to prepare the Heart Disease Prediction dataset for future machine learning modelling.

The project will examine the dataset structure and quality before addressing invalid values, duplicates, categorical variables, outliers, scaling and feature selection. These steps will produce a cleaner and more consistent dataset while ensuring that preprocessing decisions are documented and justified.

The final machine-learning-ready dataset will provide a foundation for future predictive modelling. Although the project does not itself provide a clinical diagnosis, it demonstrates how data science and systematic preprocessing can contribute to the development of future healthcare analytics solutions.

References

AnalystLab Africa. (2026). *Data Science Internship Programme: Week 2 – Feature engineering & data preprocessing.* Assignment brief.

Bhatt, C. M., Patel, P., Ghetia, T., & Mazzeo, P. L. (2023). *Effective heart disease prediction using machine learning techniques.* Algorithms, 16(2), 88. https://doi.org/10.3390/a16020088

Centers for Disease Control and Prevention. (2025). *Heart disease facts.* U.S. Department of Health and Human Services. https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html

fedesoriano. (2021, September). *Heart failure prediction dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). *Scikit-learn: Machine learning in Python.* Journal of Machine Learning Research, 12, 2825–2830. https://scikit-learn.org/stable/modules/preprocessing.html

World Health Organisation. (2025, July 31). *Cardiovascular diseases (CVDs).* https://www.who.int/en/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

WHO CVD Risk Chart Working Group. (2019). *World Health Organisation cardiovascular disease risk charts: Revised models to estimate risk in 21 global regions.* The Lancet Global Health, 7(10), e1332–e1345. https://pmc.ncbi.nlm.nih.gov/articles/PMC7025029/